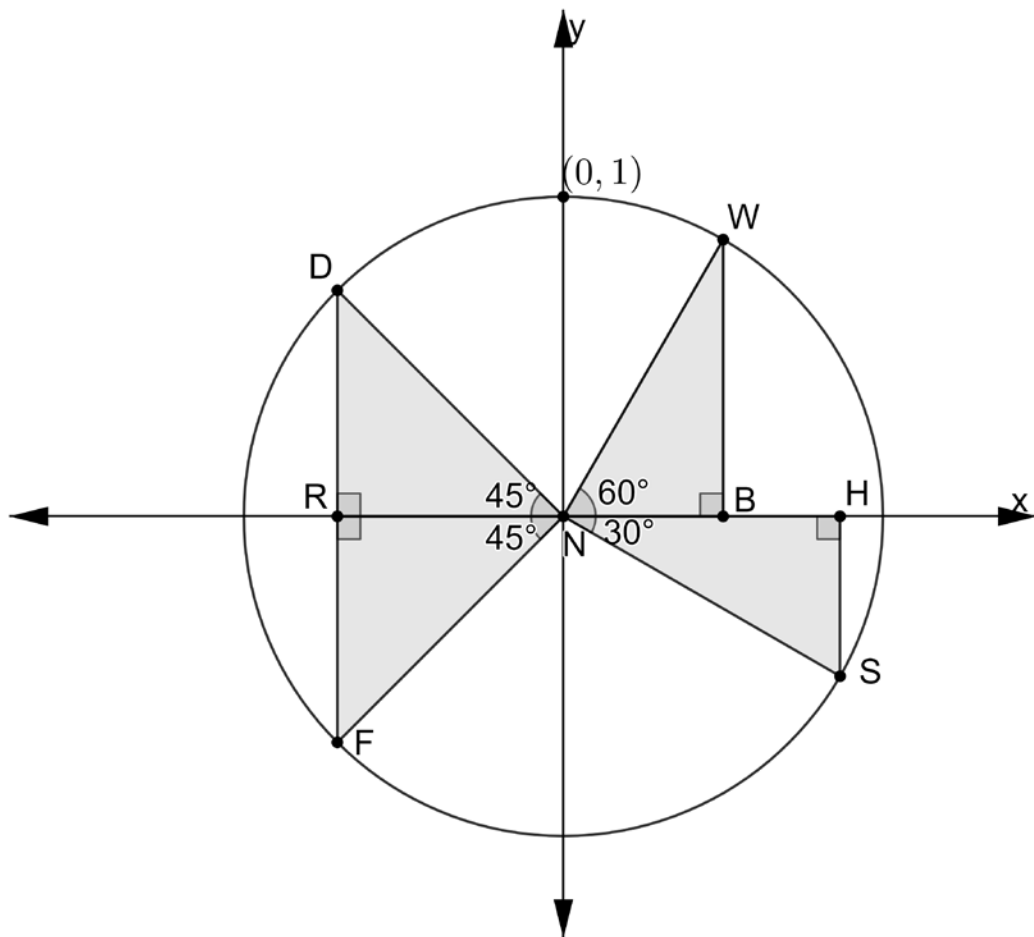


Circle  $N$  and right triangles  $NWB$ ,  $NHS$ ,  $DNR$ , and  $RFN$  are shown on the coordinate plane.



1. What is the length of the radius of circle  $N$ ?

1

Complete the table.

|    | Segment         | Length |
|----|-----------------|--------|
| 2. | $\overline{NW}$ | 1      |
| 3. | $\overline{DN}$ | 1      |

Complete the table.

|     | Point | Coordinates                                             |
|-----|-------|---------------------------------------------------------|
| 4.  | $B$   | $\left(\frac{1}{2}, 0\right)$                           |
| 5.  | $H$   | $\left(\frac{\sqrt{3}}{2}, 0\right)$                    |
| 6.  | $R$   | $\left(\frac{-\sqrt{2}}{2}, 0\right)$                   |
| 7.  | $W$   | $\left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$          |
| 8.  | $D$   | $\left(\frac{-\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$  |
| 9.  | $F$   | $\left(\frac{-\sqrt{2}}{2}, \frac{-\sqrt{2}}{2}\right)$ |
| 10. | $S$   | $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$         |